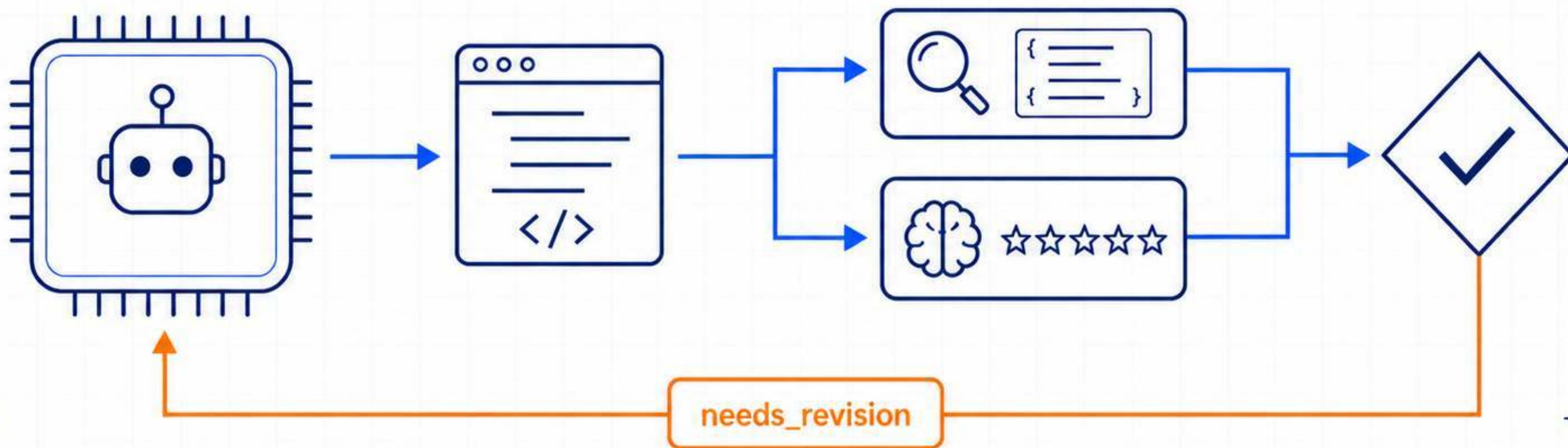


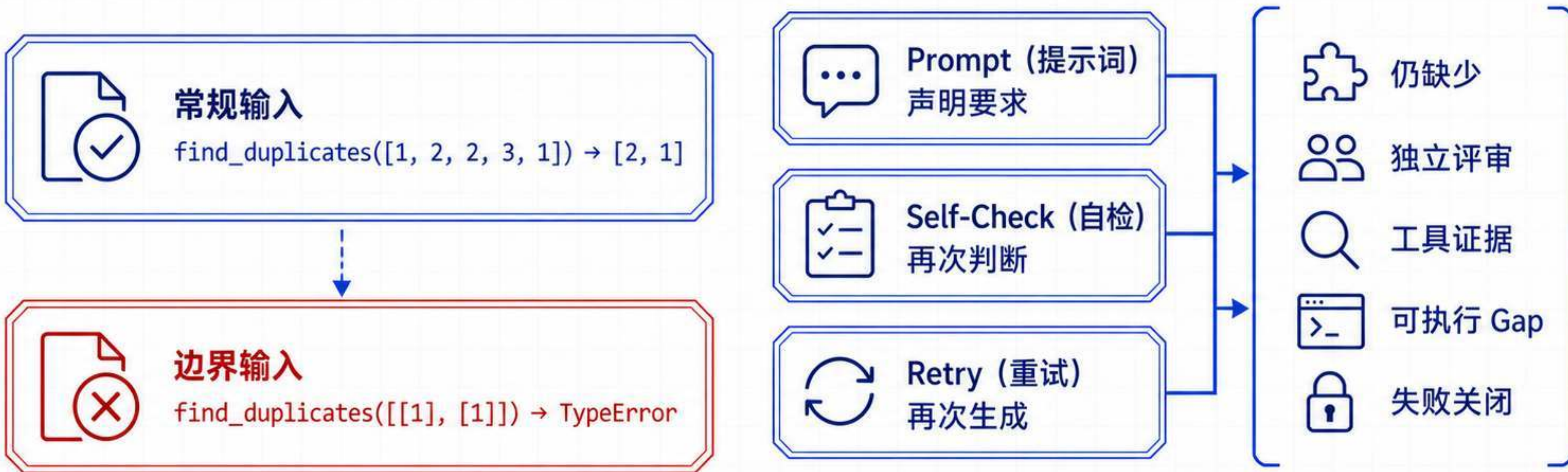
Deep Agents 实战

第 17 讲：Rubric Middleware（评分量规中间件）



生成结束，不等于通过验收

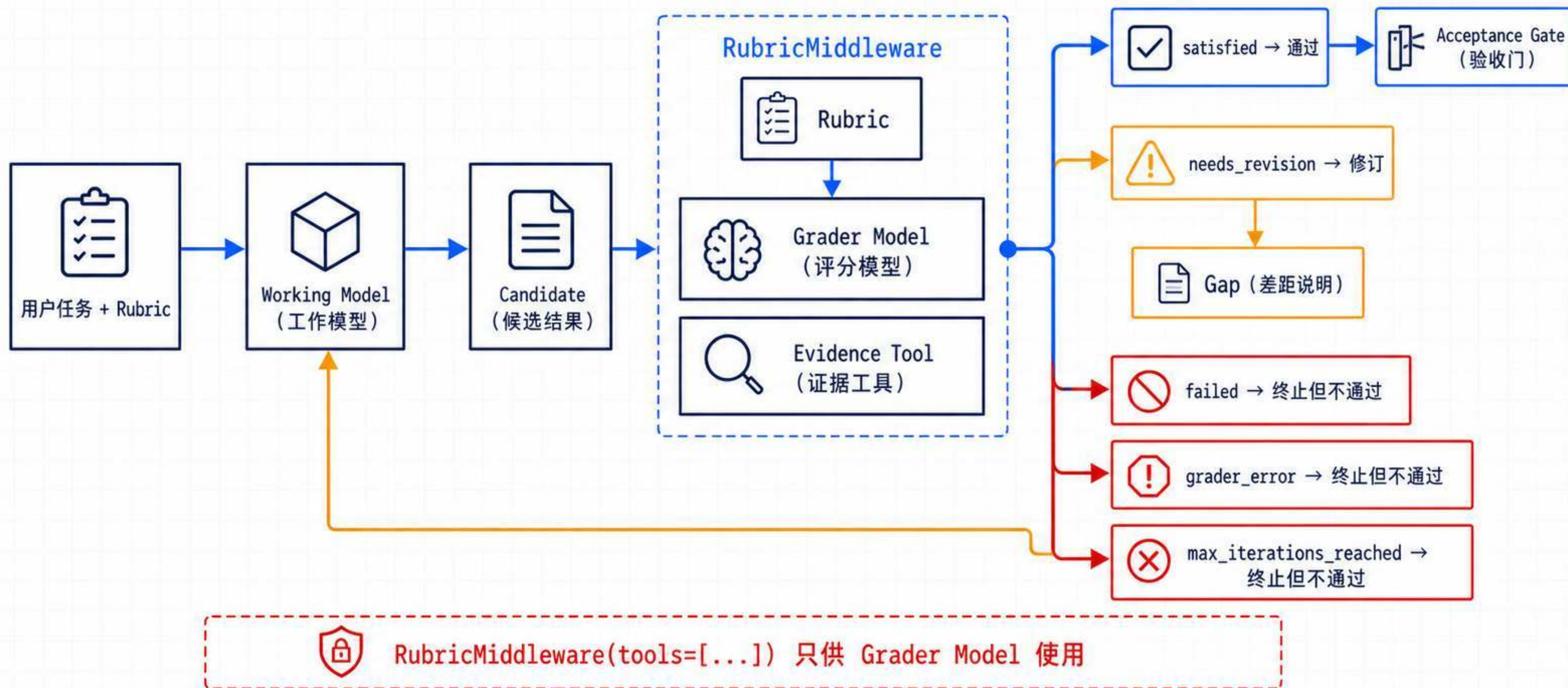
一个完整答案，仍可能在边界输入上失败



生成结束 ≠ 验收通过

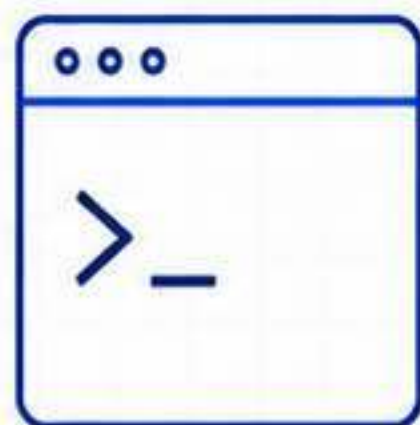
Rubric Middleware 在自然停止后接管验收

只有 needs_revision 会回环，只有 satisfied 可以放行



先安装依赖，再分开配置两个模型

工作模型负责任务，评分模型负责验收



```
uv add "deepagents==0.7.1" langchain-openai
```

```
import os
from langchain.chat_models import init_chat_model

working_model = init_chat_model(
    os.environ["WORKING_MODEL"]
)
grader_model = init_chat_model(
    os.environ.get(
        "GRADER_MODEL",
        os.environ["WORKING_MODEL"],
    )
)
```

WORKING_MODEL

provider:model-id

GRADER_MODEL

provider:model-id



deepagents>=0.6.5 | Beta API

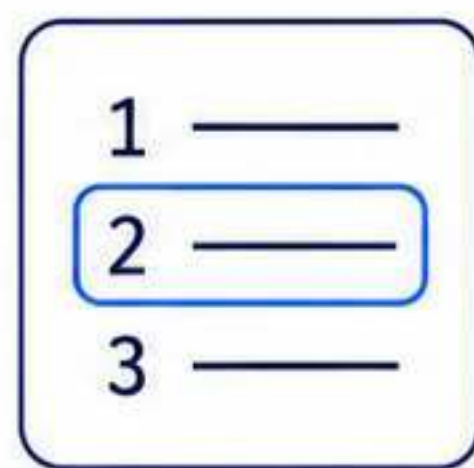
先固定 find_duplicates 的任务契约

Task、测试和 Rubric 必须使用同一种排序语义

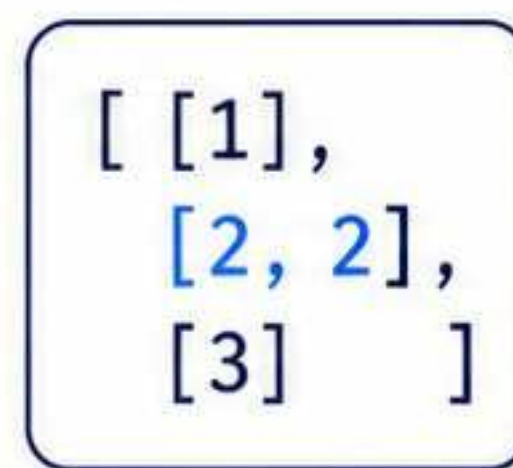
`find_duplicates(values)`



每个重复值只返回一次



按第二次出现的位置排序



支持列表等不可哈希值



不修改输入列表

把完成条件写成可取证的 Rubric

通过前必须测试最新候选，而且测试结果必须 `ok=true`

```
rubric = """
```

- Before returning satisfied, call `run_test_suite` with the latest candidate code.
- The `run_test_suite` result must contain `ok=true`.
- Each duplicated value appears exactly once.
- Result order follows the second appearance.
- Unhashable values are supported.
- The input list is not mutated.

```
""".strip()
```



可判定



可取证



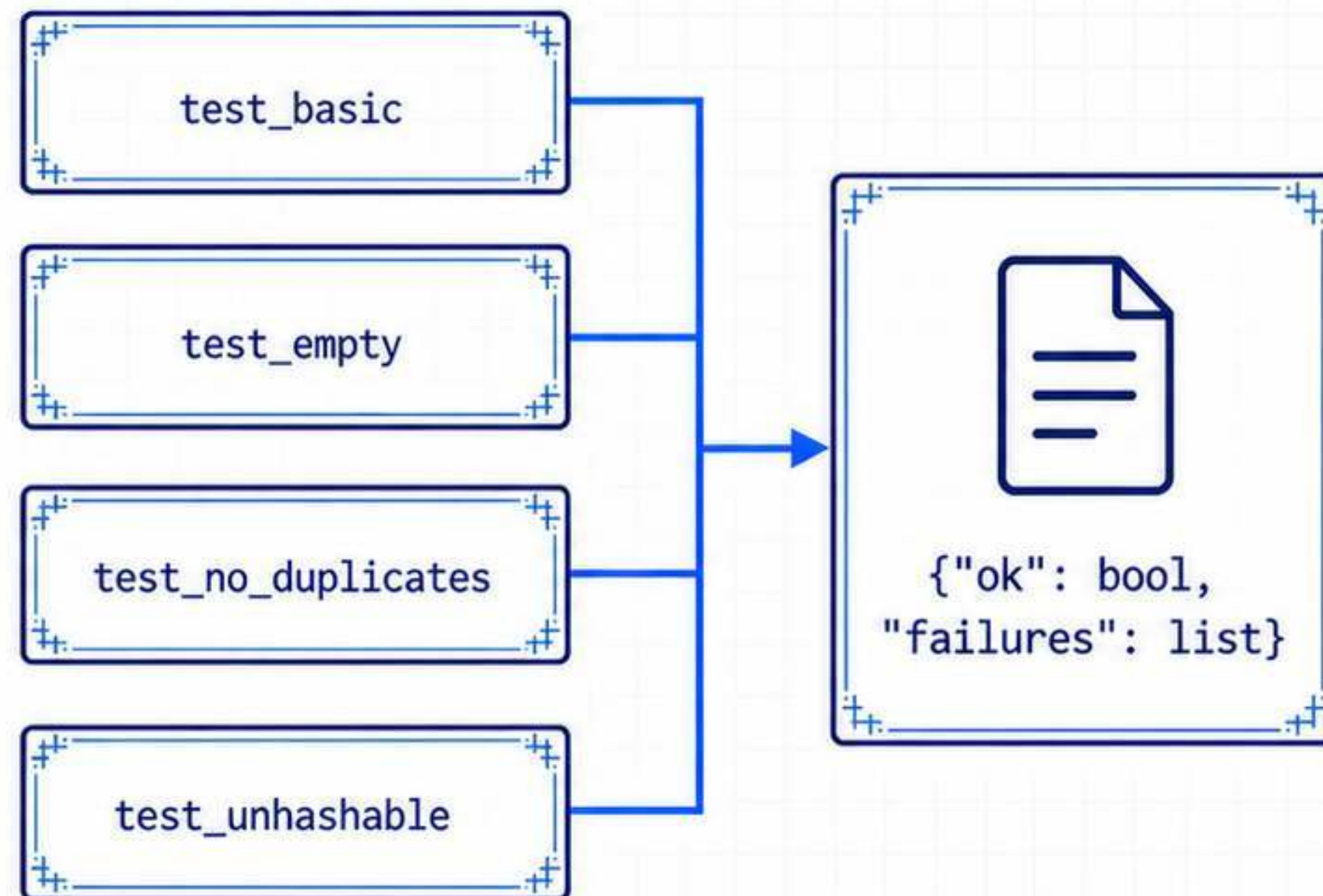
可修订

Evidence Tool 把候选源码变成测试证据

评分模型不猜代码是否正确，而是读取结构化测试结果

```
from copy import deepcopy
from langchain.tools import tool

@tool
def run_test_suite(code: str) -> dict:
    """Run behavioral tests on find_duplicates."""
    namespace = {}
    exec(code, namespace)
    fn = namespace["find_duplicates"]
    tests = [
        ("test_basic", [1, 2, 2, 3, 1], [2, 1]),
        ("test_empty", [], []),
        ("test_no_duplicates", [1, 2, 3], []),
        ("test_unhashable", [[1], [1], 2], [[1]]),
    ]
    failures = []
    for name, values, expected in tests:
        original = deepcopy(values)
        try:
            actual = fn(values)
            if actual != expected:
                failures.append(f"{name}: expected {expected}, got {actual}")
            if values != original:
                failures.append(f"{name}: input was mutated")
        except Exception as exc:
            failures.append(f"{name}: {type(exc).__name__}: {exc}")
    return {"ok": not failures, "failures": failures}
```



教学中的 `exec` 不是 Sandbox

不接模型，先证明测试工具能抓住错误

没有 API Key，也能先验证 Evidence Tool



```
def find_duplicates(values):  
    seen = set()  
    duplicates = []  
    for value in values:  
        if (  
            value in seen  
            and value not in duplicates  
        ):  
            duplicates.append(value)  
            seen.add(value)  
    return duplicates
```



```
run_test_suite.invoke({"code": bad_candidate})
```

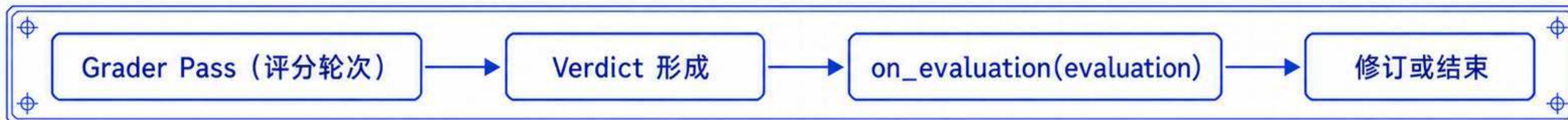
```
{  
  'ok': False,  
  "failures": [  
    "test_unhashable: TypeError: unhashable type: 'list'"  
  ]  
}
```



失败已复现

on_evaluation 在每轮评分后收到 RubricEvaluation

它记录 Grader Pass，不负责改变控制流



```
evaluations: list[RubricEvaluation] = []
```

```
def record_evaluation(evaluation):  
    run_id = evaluation["grading_run_id"]  
    evaluations.append(evaluation)
```

RubricEvaluation

grading_run_id

iteration (从 0 开始)

result

explanation

criteria[].gap



Callback (回调) 用于观察，不负责中止循环或改写 Verdict

用五个参数构造 RubricMiddleware

只有 model 必填，其余四项按需覆盖默认行为

```
rubric_middleware = RubricMiddleware(  
    model=grader_model,   
    system_prompt=(  
        "You are a strict code grader. "  
        "Always obtain current test evidence."  
    ),  
    tools=[run_test_suite],  
    max_iterations=3,  
    on_evaluation=record_evaluation,  
)
```

1 model: 必填 · 谁来评分

2 system_prompt: 可选 · 怎样评分

3 tools: 可选 · 怎样取证

4 max_iterations: 可选 · 默认 3

5 on_evaluation: 可选 · 记录结论

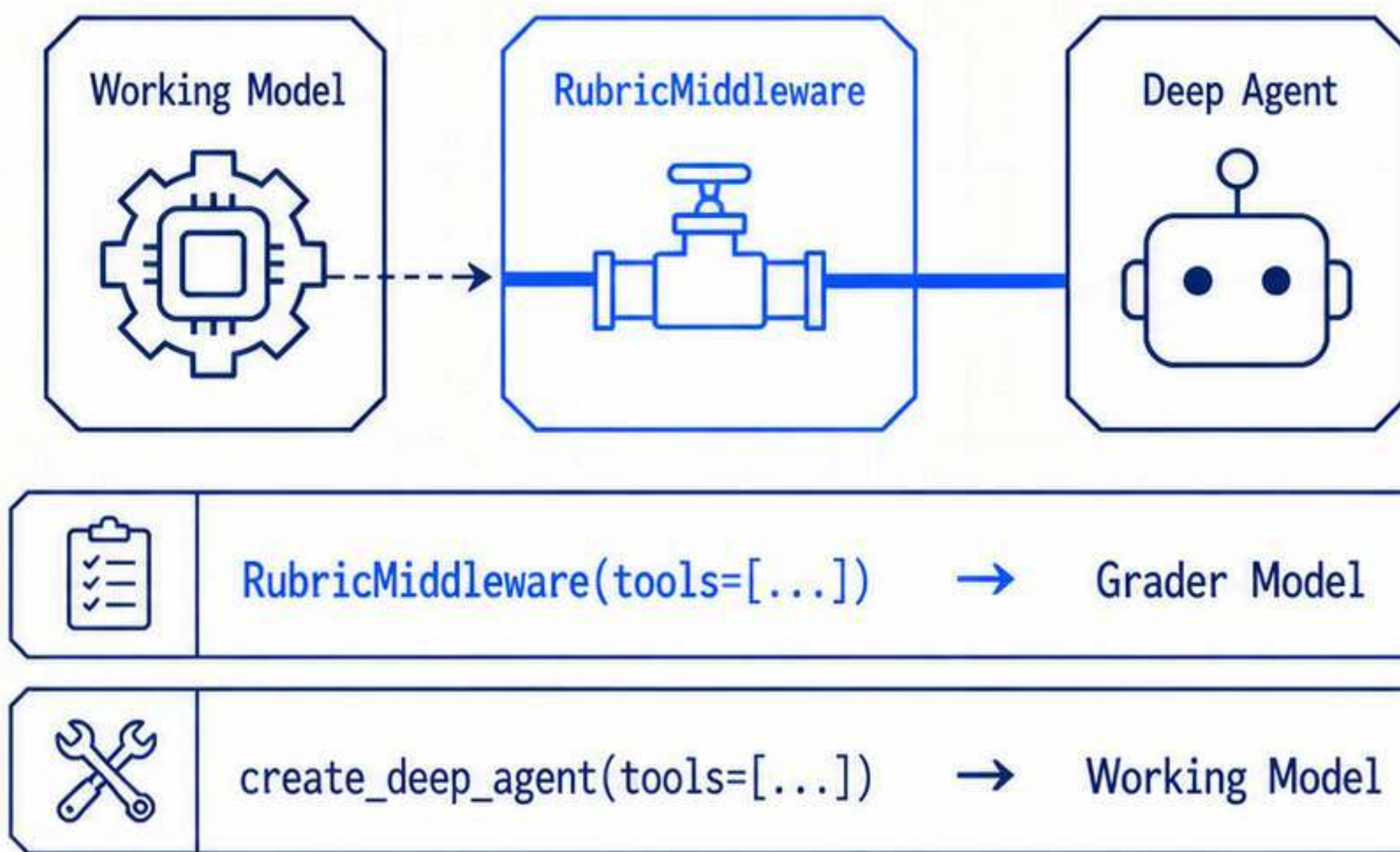


tools=[run_test_suite] 只供 Grader Model 使用

把 Middleware 挂进 Agent 才会生效

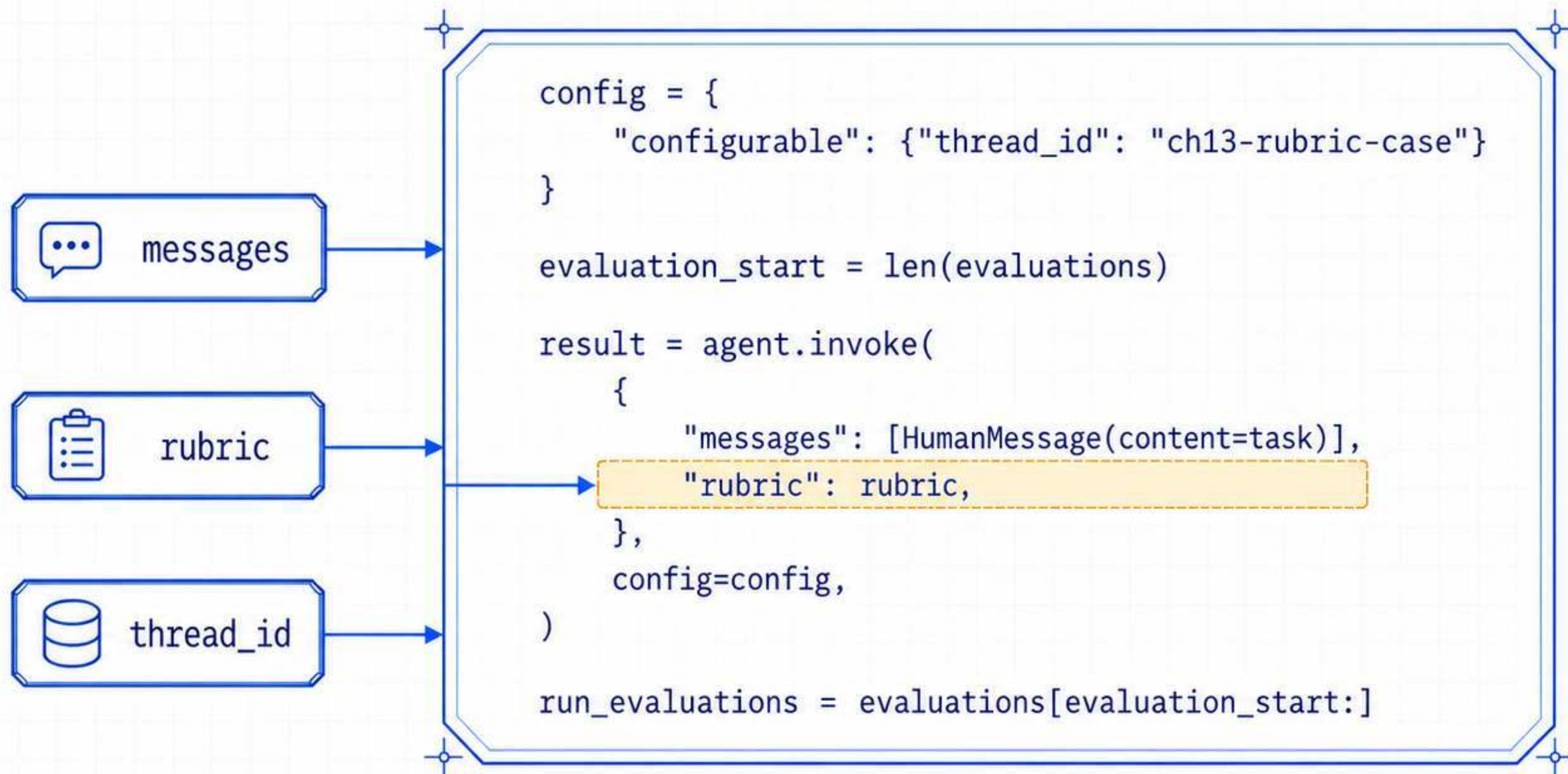
创建实例还不够，真正的挂载点是 `middleware=[rubric_middleware]`

```
agent = create_deep_agent(  
    model=working_model,  
    system_prompt=(  
        "Return only executable Python source."  
        "Revise when the grader reports a gap."  
    ),  
    middleware=[rubric_middleware],  
    checkpointer=InMemorySaver(),  
)
```



首次 invoke 显式传入 rubric, 启动评分循环

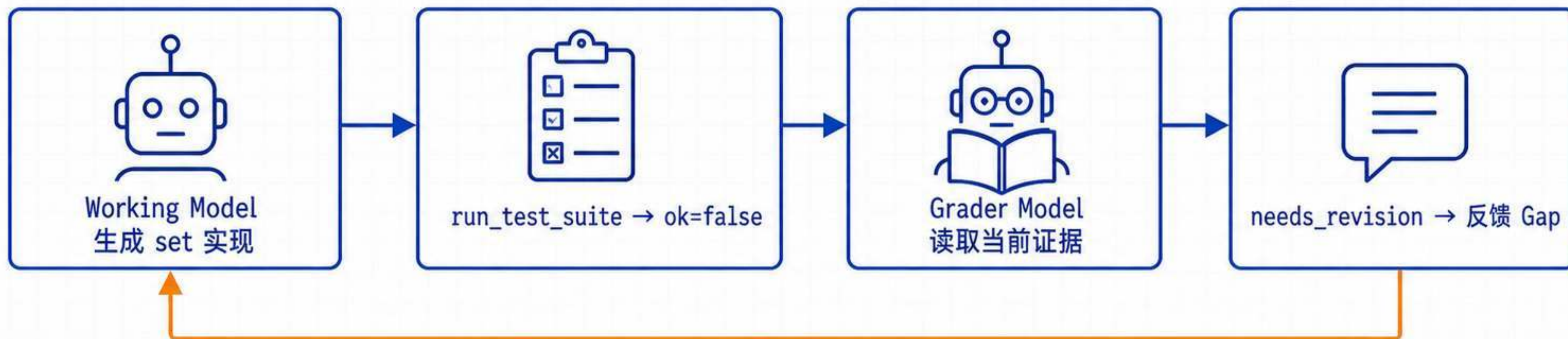
同一 thread_id 可延续已有 Rubric; 新任务使用新的 thread_id



首次调用或新任务 → 显式传入非空 rubric

第一轮失败后，needs_revision 会触发修订

真正回到 Working Model 的是可执行 Gap



```
代表性运行
iteration 0: needs_revision
gap: Unhashable values are supported
- test_unhashable raised TypeError
```



避免依赖 Hash，改用相等性比较



真实模型也可能第一轮直接通过

修订后，证据必须与当前候选对应

本例重新运行 `run_test_suite`，确认不可哈希输入已经通过

旧候选 + 旧证据

```
seen = set()
seen.add(value)
```



```
values = [[1], [1]]
test_unhashable → TypeError
旧证据不能证明新候选正确
```

当前候选 + 当前证据

```
def contains(items, target):
    return any(item == target for item in items)
```

```
seen = []
if contains(seen, value):
    duplicates.append(value)
```

`run_test_suite`

```
{'ok': True, 'failures': []}
```

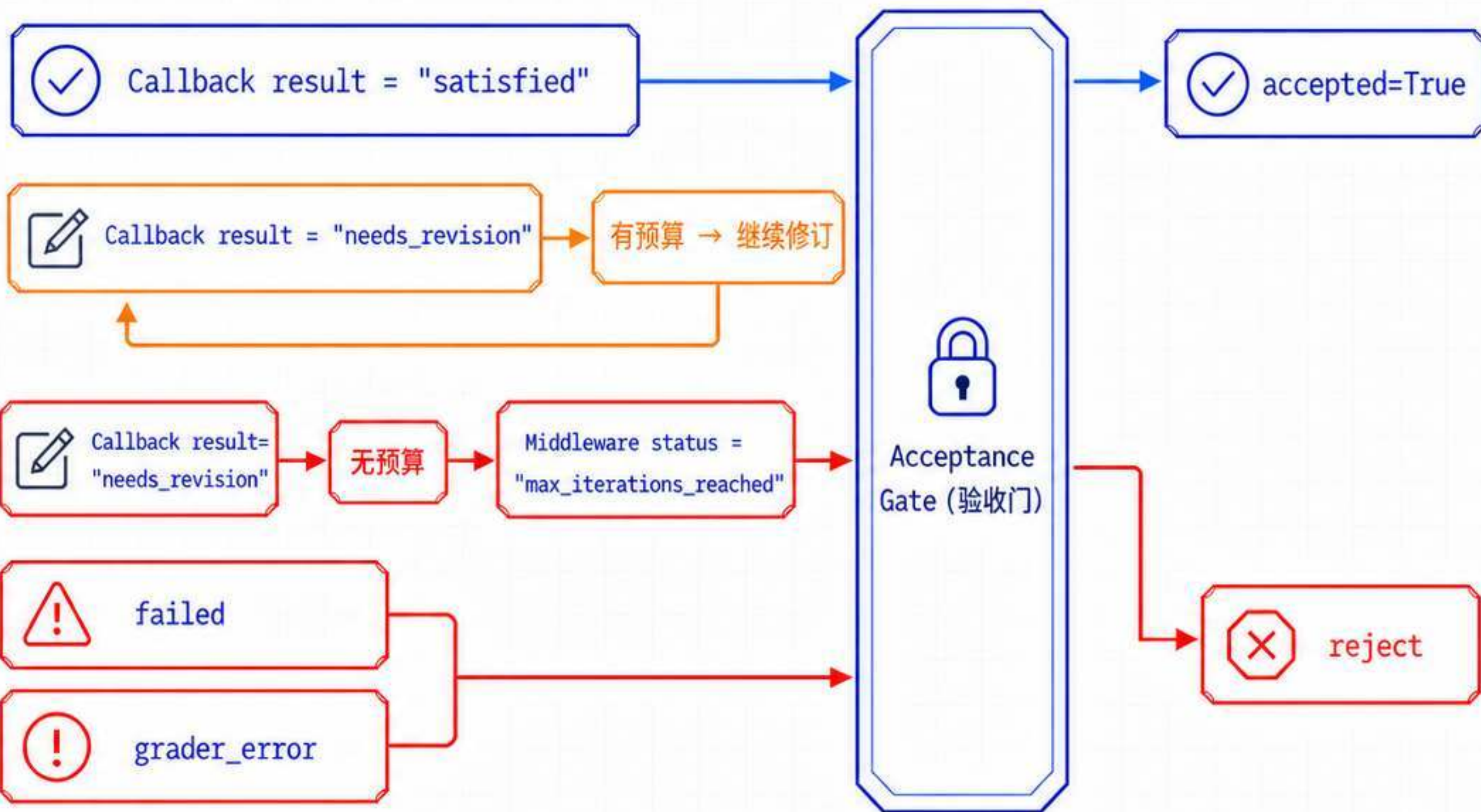
iteration 1: satisfied



Callback 记录本轮 Verdict, 验收门决定是否放行

needs_revision 可能耗尽预算; 只有 satisfied 才能 accepted=True

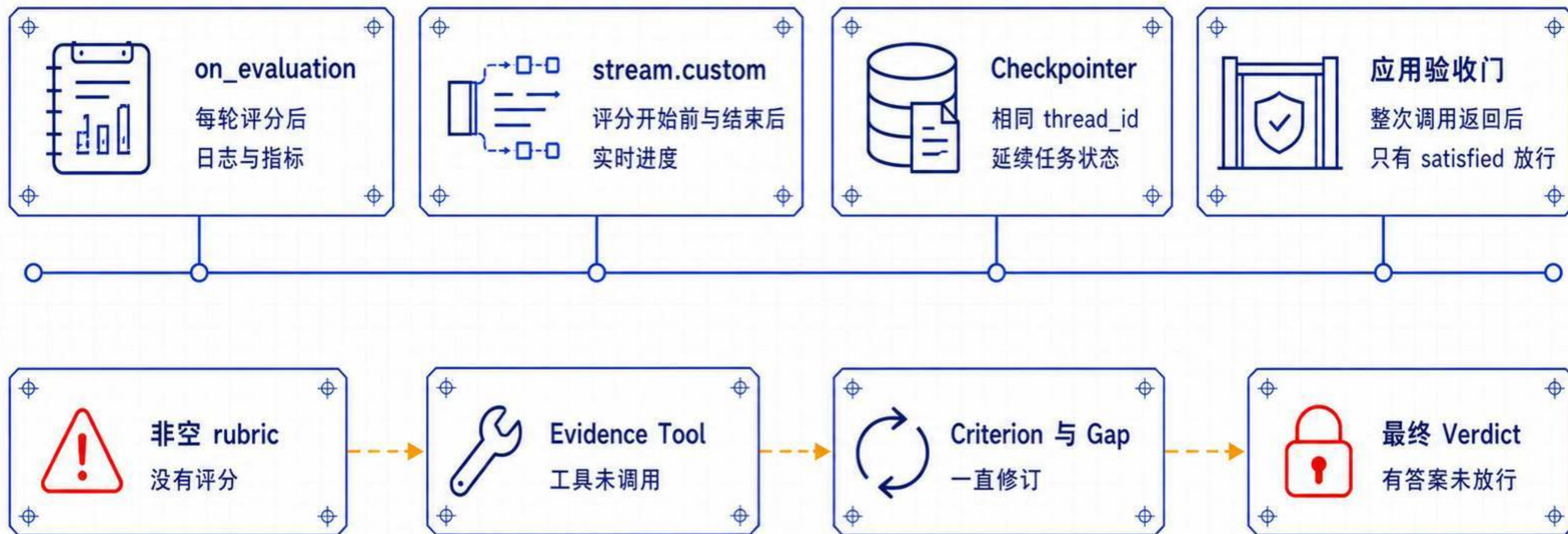
```
accepted = (  
    final_evaluation is not None  
    and final_evaluation["result"] == "satisfied"  
)
```



达到上限时, Callback 中的 needs_revision 不会被改写

观察、状态延续与放行各有独立入口

Callback 看结果，事件流看进度，Checkpoint 续状态，验收门做决定



用 AgentSeek Template 体验 Rubric Middleware

推荐先跑无密钥演示，再连接真实模型观察完整评分回环

🔧 安装与启动

01 先安装 AgentSeek

```
$ uv tool install agentseek
```

02 创建 langchain/rubric 模板

```
$ agentseek create langchain/rubric --checkout main  
$ cd rubric_lab
```

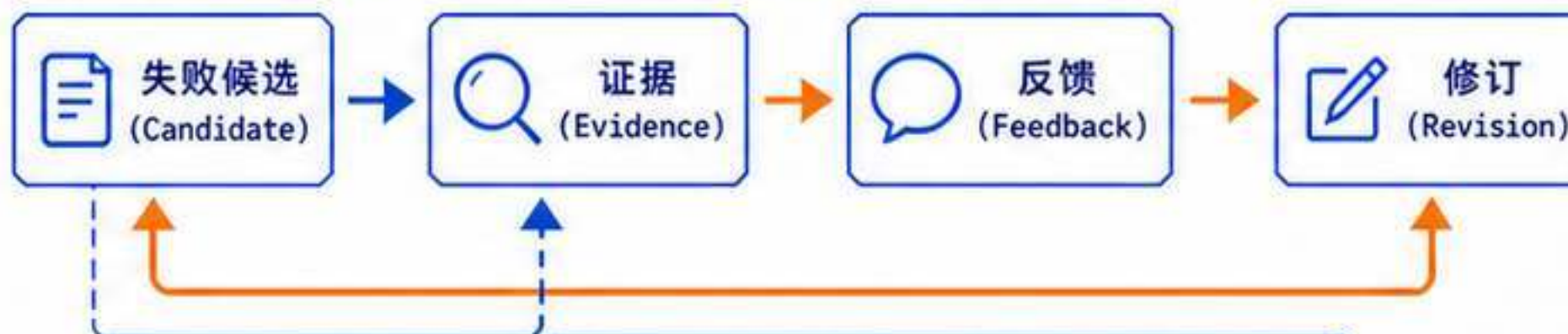
03 无密钥验证并启动

```
$ cp .env.example .env  
$ agentseek task sync  
$ agentseek task frontend  
$ agentseek task rubric-smoke  
$ agentseek dev
```

🧪 模板核心内容

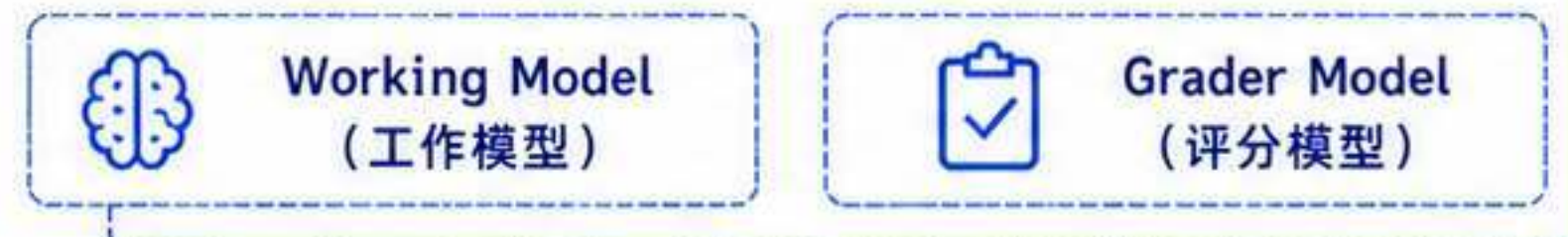
1 Guided Demo (引导演示)

无需模型密钥，复现失败候选 → 证据 → 反馈 → 修订 → 通过



2 Live Model (真实模型)

分别配置 Working Model (工作模型) 与 Grader Model (评分模型)



3 Evidence Loop (证据回环)

Evidence Tool (证据工具) 绑定当前候选，评分结果可追溯



4 Acceptance Gate (验收门禁)



受限子进程不是 Isolated Sandbox ; Live Model 凭据只放在服务端 .env